

Recursive Green's function method for graphene: A review

Submitted By

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Abstract

Graphene is significant material of interest with distinct electronic characteristics. The recursive Green's function approach is an effective and robust method for computation of electronic transport quantities in mesoscopic systems. In this work, we use this method to investigate the transport characteristics of single layer graphene in the ballistic conductor regime. It is found to be particularly effective for larger systems, enabling computation of relevant quantities such as conductance, local density of states, and current distributions. We also comment on the usability of this method for systems with disorder and interaction, and with finite temperature.

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Chapter 1

Introduction: Mesoscopic Transport

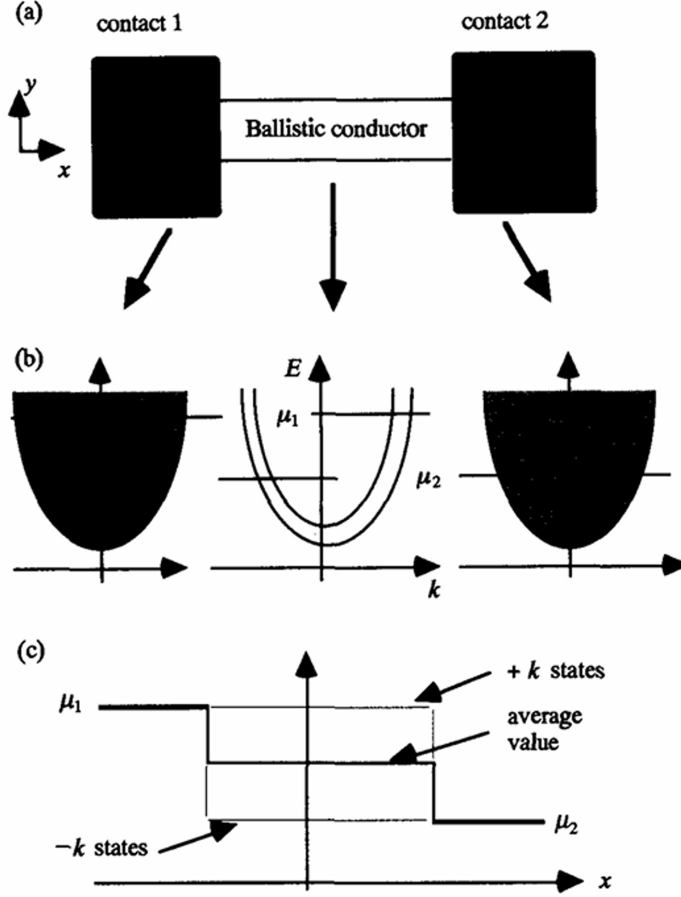
Small conductors whose dimensions are intermediate between the microscopic and the macroscopic are called mesoscopic. They are much larger than microscopic objects like atoms, but not large enough to be ‘ohmic’. A conductor usually shows ohmic behavior if its dimensions are much larger than each of three characteristic length scales: (1) the de Broglie wavelength, which is related to the kinetic energy of the electrons, (2) the mean free path and (3) the phaserelaxation length, which is the distance that an electron travels before its initial phase is destroyed. These length scales vary widely from one material to another and are also strongly affected by temperature, magnetic field etc.

1.1 Conductance from transmission

1.1.1 Conductance in a Ballistic conductor

Consider a conductor stretched between two contact pads as shown in Fig 1.1. Its conductance is expected to follow Ohm’s law, represented as $G = \sigma W/L$, with σ being a material constant. However, as the length (L) of the conductor decreases, experiments show that conductance approaches a limiting value G_c instead of increasing indefinitely, which is unrelated to quantum mechanics. This observed resistance is attributed to the interface between the conductor and the dissimilar contact materials, known as contact resistance (G_c^{-1}). While a ballistic conductor ideally has no resistance, the need for current redistribution at this interface introduces resistance. Although one could theoretically eliminate contact resistance by matching materials, it would compromise the measurement’s validity, as contacts must be significantly more conductive than the conductor.

To assess G_c^{-1} , we examine current flow through a ballistic conductor under an applied bias, assuming ‘reflectionless’ contacts, allowing electrons to transition into contacts without reflections. Numerical analyses confirm that when an electron exits a narrow conductor into a wider contact, reflection probability remains low as long as energy is not close to the band’s bottom. However, reflections increase significantly when traveling back from the contact to the conductor, asserting the concept of ‘reflectionless’ contacts.



(a) A ballistic conductor connected to two wide contacts. (b) Inside the wide contacts there is a very high density of transverse modes (indicated by shading) and both $+k$ and $-k$ states have nearly the same quasi-Fermi level. In the narrow ballistic conductor there are only a few modes and the $+k$ and $-k$ states have different quasi-Fermi levels. (c) Variation of the electrochemical potential from one contact to the other. Note that k stands for k_x .

Fig. 1.1

In the case of 'reflectionless' contacts, $+k$ states in the conductor are filled solely by electrons from the left contact, while $-k$ states are exclusively occupied by electrons from the right contact. This occurs because right-contact electrons fill $-k$ states without reflecting into the left contact, whereas left-contact electrons occupy $+k$ states without reflecting into the right contact. The quasi-Fermi level F^+ for the $+k$ states remains constant even under applied bias.

When both contacts are at the same potential, the Fermi level for the $+k$ states corresponds to this potential. Changing the potential at the right contact has no impact on the quasi-Fermi level of $+k$ states since no right-contact electrons can reach these states. Similarly, the quasi-Fermi level F^- for the $-k$ states in the right contact is always equal to the potential there. Therefore, at low temperatures, the total current is determined by the $+k$ states situated between the two contacts. This understanding underscores the independence of quasi-Fermi levels on potential changes in unrelated contacts, which is crucial for analyzing electron transport in such systems.

The states in the narrow conductor belong to different transverse modes or subbands. Each mode has a dispersion relation $E(N, k)$ as sketched in Fig 1b with a cut-off energy ε_N below which it cannot propagate. The number of transverse modes at an energy E is obtained by counting the number of modes $M(E) = \sum_N \theta(E - \varepsilon_N)$ having cut-off energies smaller than E .

We can evaluate the current carried by each transverse mode separately and add them up.

Consider a single transverse mode whose $+k$ states are occupied according to some function $f^+(E)$. A uniform electron gas with n electrons per unit length moving with a velocity v carries a current equal to env . Since the electron density associated with a single k -state in a conductor of length L is $(1/L)$, the current I^+ carried by the $+k$ states is described as,

$$I^+ = \frac{e}{L} \sum_k v f^+(E) = \frac{e}{L} \sum_k \frac{1}{\hbar} \frac{\partial E}{\partial k} v f^+(E) \quad (1.1)$$

Assuming periodic boundary conditions we obtain,

$$I^+ = \int_{-\infty}^{+\infty} f^+(E) dE \quad (1.2)$$

where ε is the cut-off energy of the waveguide mode. This is extended to multi-moded waveguides and the current, I^+ , carried by the $+k$ states in a conductor is given by,

$$I^+ = \int_{-\infty}^{+\infty} f^+(E) M(E) dE \quad (1.3)$$

Note that this is a general result independent of the actual dispersion relation $E(k)$ of the waveguide: the current carried per mode per unit energy by an occupied state is equal to $2|e|/\hbar$.

Assuming that the number of modes M is constant over the energy range $\mu_1 > E > \mu_2$, we can write,

$$I = \frac{2e^2}{h} M \frac{\mu_1 - \mu_2}{e} \implies G_C = \frac{2e^2}{h} M \quad (1.4)$$

so that the contact resistance (which is the resistance of a ballistic waveguide) is given by

$$G_C^{-1} = \frac{(\mu_1 - \mu_2)/e}{I} = \frac{h}{2e^2 M} \quad (1.5)$$

To calculate the number of modes $M(E)$ we need to know the cut-off energies for the different modes ε_N , which depend on the confining potential $U(y)$ and the magnetic field. However, for wide conductors in zero magnetic field the precise nature of the confining potential is not important. The number of propagating modes can be written as,

$$M = \text{Int}\left[\frac{k_f W}{\pi}\right] = \text{Int}\left[\frac{2W}{\lambda_f}\right] \quad (1.6)$$

where $\text{Int}(x)$ represents the integer that is just smaller than x .

The contact resistance can be measured directly using point contacts to create a constriction (much shorter than a mean free path) in a conductor. This was done in the late 1960s in metals. In metals the Fermi wavelength is extremely short, of the order of the distance between atoms. Consequently the number of modes M is quite large and the contact resistance is relatively small.

In semiconductors on the other hand, the Fermi wavelength is typically $\sim 30\text{nm}$ so that the factor $k_f W$ can easily be made quite small. The first experiment on semiconductors was reported independently by two groups in 1988. As the width W of the constriction was reduced the conductance went down in discrete steps each of height $(2e^2/h)$. This is in agreement with our

analysis since $M \in \mathbb{Z}$. Although the width of a conductor changes continuously, the number of modes changes in discrete steps. This discreteness is not evident if the conductor is many thousands of wavelengths wide, since a very small fractional change in W changes M by many integers.

Landauer Formula

Consider a conductor connected to two large contacts by two leads as shown in Fig 1.2.

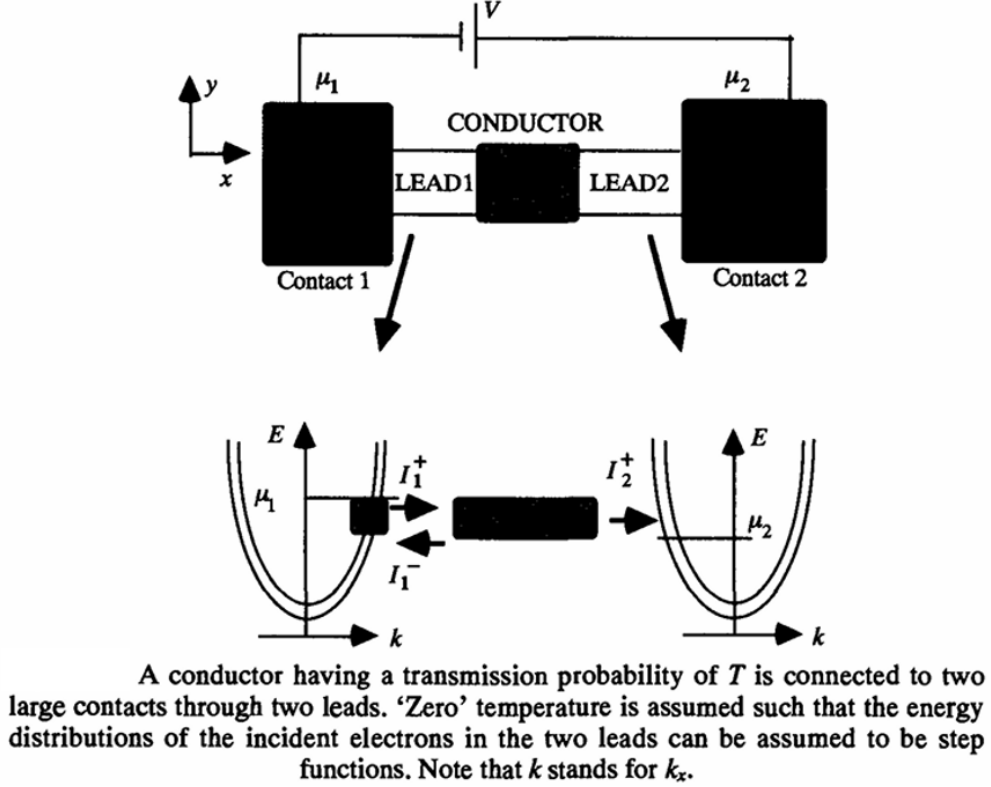


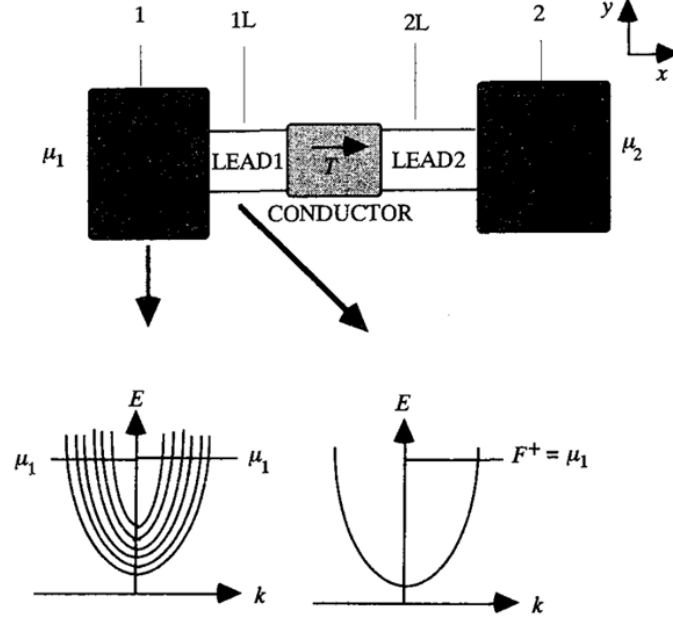
Fig. 1.2

The leads are assumed to be ballistic conductors, each having M transverse modes. T is the average probability that an electron injected in lead 1 will transmit to lead 2. We assume that electrons can exit from the conductor into the contacts without any reflection (that is, the contacts are 'reflectionless'). Then the $+k_x$ states in lead 1 are occupied only by electrons coming in from the left contact and hence these states must have an electrochemical potential of μ_1 . Similarly the $-k_x$ states in lead 2 are occupied only by electrons coming in from the right contact and hence must have an electrochemical potential of μ_2 . Assuming zero temperature, the Landauer formula is given by,

$$G = \frac{2e^2}{h} MT \quad (1.7)$$

As discussed, the conductance expressed in 1.7 includes contact resistance, representing total conductance between two specific planes deep within the contacts, labeled '1' and '2' in Fig 1.3, as opposed to planes in the leads, labeled '1L' and '2L'. A pertinent question arises: should one compute the transmission probabilities between '1' and '2' instead of between '1L' and '2L'? The answer is negative, provided the contacts are 'reflectionless'; in this situation, electrons enter the contacts from the leads with minimal reflection probability.

For ballistic conductors, calculating the transmission probability between points $1L$ and $2L$ is straightforward, yielding $T = 1$ for all modes, which in turn allows the conductance to equal $(2e^2/h)M$, where M counts the modes in the constriction. Conversely, evaluating the transmission between planes '1' and '2' in the wider contacts is complex, resulting in a much larger number of modes but a smaller average transmission probability per mode. However, this complexity does not affect the resultant conductance value, which remains the same $((2e^2/h)M)$. The conductance measured between planes '1' and '2' can adequately be derived from evaluating the transmission between ' $1L$ ' and ' $2L$ '.



In the wide contacts the current carried per mode is small so that $+k$ and $-k$ states have about the same quasi-Fermi level. If the contacts are 'reflectionless' then the $+k$ states in lead 1 are in equilibrium with contact 1 and have a quasi-Fermi level μ_1 while the $-k$ states in lead 2 (not shown) are in equilibrium with contact 2.

Fig. 1.3: Caption

This emphasizes the significance of the incoming electrons' energy distribution at the lead, while the outgoing electrons' distribution is inconsequential. We can assert that with 'reflectionless' contacts, incoming states in each lead achieve thermal equilibrium with their respective contacts. This allows for the straightforward application of the Landauer formula when calculating transmission between two planes ' $1L$ ' and ' $2L$ ' within the leads. Numerical findings suggest that, provided the energy level is not excessively close to the band minima, electrons can transition from a narrow lead to a wide contact with minimal reflection. In such reflectionless contacts, electrons traveling from the right contact to the conductor populate $-k$ states in lead 1, while $+k$ states remain attributed solely to electrons from the left contact, thereby ensuring equilibrium under bias conditions. Consequently, the quasi-Fermi level F^+ for $+k$ states in lead 1 aligns with μ_1 , and similarly, F^+ for $-k$ states in lead 2 corresponds to μ_1 .

Source of resistance

The expression for conductance (1.7) indicates that scatterers give rise to resistance by reducing the transmission probability (apart from the contact resistance). For the given setup (reflectionless contacts) and neglecting the interference effects and treating electrons semi-classically,

the $+k$ states to the left of the scatterer are occupied only by electrons coming in from the left contact. Hence these states have the same electrochemical potential as the left contact μ_1 , and similarly for $-k$ states to the right with μ_2 . The partial occupation of $-k$ states before the scatterer or the $+k$ states after the scatterer (in energy range $\mu_1 > E > \mu_2$) depends on transmission probability T .

These distributions are highly non-equilibrium distributions and are only valid very near the scatterer. If we move more than a few energy relaxation lengths away from the scatterer, the electrons will settle down to lower energies and a Fermi distribution will be established (we assume that the energy relaxation processes help establish equilibrium among the $+k$ states and among the $-k$ states but do not cause any transfer of electrons between the two groups). This evolution of the electron energy distribution requires heat dissipation, which occurs near (over $\sim$ energy relaxation length) the scatterer. Thus, the potential drop (associated to resistance due to scattering) occurs right across the scatterer, and average energy of the current changes slowly over the relaxation length. The sharp change in quasi-Fermi energy across the scatterer leads to the formation of resistivity dipoles at the scatterer. Therefore, even a ‘homogeneous conductor with a uniform conductivity’ is really extremely inhomogeneous on a mesoscopic scale.

1.1.2 Probe measurement

The standard method for measuring resistance in large conductors employs a multi-terminal Hall bridge, with current entering and exiting through designated terminals while additional floating terminals act as voltage probes to detect local electrochemical potentials. When a Hall bridge is created with two voltage probes positioned near a scatterer, the varying electrochemical potentials for $+k$ and $-k$ states come into play. However, three primary issues arise with mesoscopic probes.

Firstly, *these probes can be invasive*, altering the measurement environment. In macroscopic cases, probes minimally impact the current, but in smaller conductors, they may contribute significantly to scattering and resistance. Despite this, there is no inherent restriction on the coupling strength of voltage probes, and research using weakly coupled scanning tunneling probes has shown potential for observing resistivity dipoles around scatterers.

Secondly, *mesoscopic probes are not typically identical*, leading to varying coupling to $+k$ and $-k$ states. For instance, if a probe is angled towards the right, it may better couple to $+k$ states, registering a potential closer to that state, while a probe angled to the left will show a stronger coupling to $-k$ states. This variance in probe orientation significantly influences measurement outcomes. The measured resistance in conductors with $T > 0.5$ can even be negative. In practical applications, control over microscopic potential profiles affecting probe coupling to $+k$ and $-k$ states is limited. We anticipate measuring the resistance described by 1.7 only when voltage probes couple identically. For lengthy conductors, the transmission probability is significantly less than one, resulting in minimal distinction between 1 and $(1 - 2T)$.

Lastly, *mesoscopic measurements are greatly influenced by quantum interference* unless the probes are far from the scatterer compared to the phase-relaxation length. For instance, with a strongly reflecting scatterer where T is very low, electrochemical potentials adjacent to the scatterer are nearly equal, yet the left probe may measure values between zero and one due to quantum interference. This occurs because destructive interference can prevent the probe from detecting electrons from the left reservoir, causing it to appear closer to the right reservoir’s potential. 1.7 applies to four-terminal resistance primarily when interference effects are minimal or accounted for. This highlights the limitations of semiclassical methods

previously discussed, as the analyzed distribution functions and quasi-Fermi energies do not encompass the strong correlations present in phase-coherent conductors, which is especially relevant in scenarios like p-n junctions where electron and hole energies differ significantly.

Buttiker formula

Since there is no qualitative difference between the current and voltage probes, Buttiker extended the two-terminal linear response formula, $I = \frac{2e}{h}\bar{T}[\mu_1 - \mu_2]$, by summing over all the terminals (indexed by p and q) as (treating all the probes equally),

$$I_p = \frac{2e}{h} \sum_q [\bar{T}_{q \leftarrow p} \mu_p - \bar{T}_{p \leftarrow q} \mu_q] \quad (1.8)$$

We can rewrite,

$$I_p = \frac{2e}{h} \sum_q [G_{qp} V_p - G_{pq} V_q] \quad (1.9)$$

where,

$$G_{pq} = \frac{2e^2}{h} \bar{T}_{p \leftarrow q} \quad (1.10)$$

1.1.3 Finite temperature, bias, and non-coherent transport

So far, we have considered the assumption of zero temperature so that transport occurred only from contact 1 to 2 and not from 2 to 1 (for a two-terminal setup). We also assumed that the current was carried by a single energy channel around the Fermi energy. In general, transport takes place through multiple energy channels in the energy range $\mu_1 + (\sim k_B T) > E > \mu_2 - (\sim k_B T)$ and each channel can have a different transmission $\bar{T}$, as shown in Fig 1.4. 4. Under these conditions, the net current is given by,

$$i(E) = \frac{2e}{h} [\bar{T}(E) f_1(E) - \bar{T}'(E) f_2(E)] \quad (1.11)$$

In general, $\bar{T}(E) \neq \bar{T}'(E)$. The equality holds for no inelastic scattering (from one energy to another) inside the device, even in the presence of magnetic fields. Also, the two functions are equal at equilibrium (there is no current at equilibrium), however, as we move away from equilibrium, the applied bias could change the two transmission functions and make them unequal.

For small deviations from this equilibrium state, the current is proportional to the applied bias. We can write,

$$\delta I = \frac{2e}{h} \int ([\bar{T}(E)]_{eq} \delta[f_1 - f_2] + [f_1 - f_2]_{eq} \delta[\bar{T}(E)]) dE \quad (1.12)$$

Using a Taylor's series expansion and the equilibrium Fermi function $f_0(E)$ we get the (non-zero temperature) linear response as,

$$G = \frac{\delta I}{(\mu_1 - \mu_2)/e} = \frac{2e}{h} \int \bar{T}(E) \left(-\frac{\partial f_0}{\partial E} \right) dE \quad (1.13)$$

The response is linear regardless of the temperature if the transmission function $\bar{T}(E)$ is approximately constant over the energy range where transport occurs, and it can be assumed that it is not affected by bias. A general criterion for linear response is given by,

$$I = \frac{1}{e} \int_{\mu_1}^{\mu_2} \tilde{G}(E') dE' \quad (1.14)$$

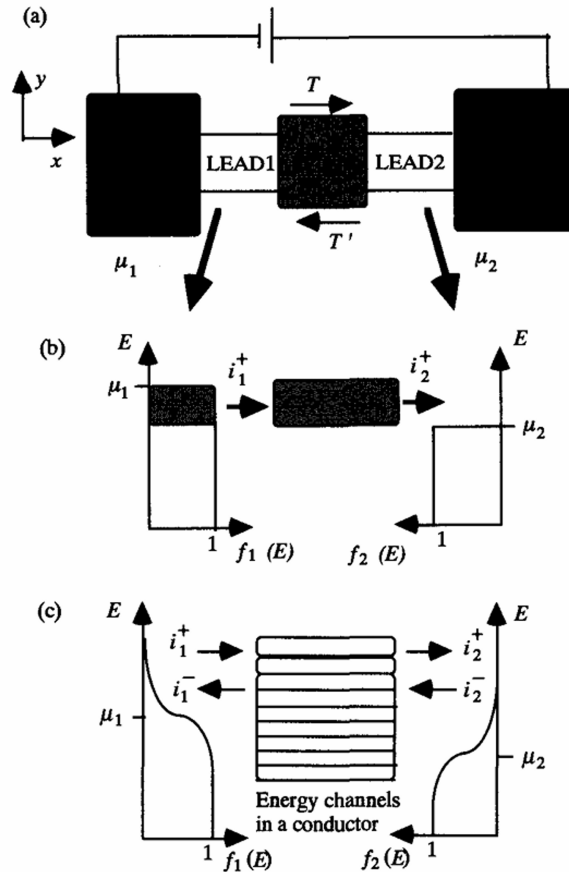
where the conductance function is defined as

$$\tilde{G}(E') = \frac{2e^2}{h} \int \bar{T}(E) F_T(E - E') dE \quad (1.15)$$

$F_T(E)$ being the thermal broadening function,

$$F_T(E) = -\frac{d}{dE} \left(\frac{1}{\exp(E/k_B T) + 1} \right) = \frac{1}{4k_B T} \text{sech}^2\left(\frac{E}{2k_B T}\right) \quad (1.16)$$

The current will respond linearly if the conductance function $\tilde{G}(E)$ is independent of energy in the energy range $\mu_1 > E > \mu_2$.



(a) A conductor is connected to two large contacts through two leads.
(b) Energy distributions of the incident electrons in the two leads at zero temperature.
(c) Energy distributions at non-zero temperatures.

Fig. 1.4

At low temperatures when the phase-relaxation time is long, quantum interference can give rise to sharp resonant structures in the transmission characteristics of conductors having

multiple sources of scattering. This is why the transmission $\bar{T}(E)$ often changes rapidly with energy (the correlation energy is small) at low temperatures in mesoscopic conductors.

At absolute zero temperature, where the thermal broadening function becomes a delta function, the response is linear only if the applied bias is significantly smaller than the correlation energy. Temperature, however, causes the sharp features to smooth out, resulting in a conductance function $\tilde{G}(E)$ that varies on the order of thermal energy $k_B T$ ($\sim 0.01 meV$ at $T = 0.1 K$). Therefore, linear response is maintained as long as the bias remains below $k_B T$, which is essential in mesoscopic experiments to remain within the linear response regime. The condition for linear response in systems with small correlation energy is given by $(\mu_1 - \mu_2) \ll k_B T$. In contrast, when correlation energy is high, the transmission function $\bar{T}$ is already smooth, making additional smoothing unnecessary and allowing for linear response at larger bias values.

As bias increases, an electric field forms within the conductor, altering $\bar{T}(E)$ and potentially leading to non-linear responses. Thus, one needs to consider the transmission as $\bar{T}(E, \mu_1, \mu_2)$. Historically, the initial applications of such formulas focused on I-V characteristics of tunneling junctions, where applied bias reduces the tunneling barrier and exponentially increases transmission. More recently, similar equations have been applied to analyze the I-V characteristics of resonant tunneling devices, emphasizing the need to factor in electric field effects on transmission, often calculated using the Poisson equation.

Scattering states

The current originates from electronic transitions between eigenstates localized in leads p and q . These transitions are impeded if the final state is already filled, prompting the inclusion of the $(1 - f)$ factors, as follows:

$$i_p(E) = \frac{2e}{h} \sum_q [\bar{T}_{qp}(E) f_p(E) (1 - f_q) - \bar{T}_{pq}(E) f_q(E) (1 - f_p)] \quad (1.17)$$

However, this approach is valid only under weak coupling conditions between the leads and the conductor. When the coupling is enhanced, lead eigenstates transition into scattering states, illustrated in Fig. 5. A scattering state features an incident wave in lead q and scattered waves across lead p , carrying current from lead q to lead p without requiring state transitions, hence negating the necessity for $(1 - f)$ factors.

To simplify, we consider one transverse mode per lead, treating each mode as a terminal. The wavefunction in lead p from a scattering state (q, k) is denoted by $(\delta_{pq} = 1; 0 \text{ otherwise})$,

$$\Psi_p(q) = \delta_{pq} \chi_p^+(y_p) \exp\{[ik^+ x_p]\} + s'_{pq} \chi_p^-(y_p) \exp\{[ik^- x_p]\} \quad (1.18)$$

with χ_p^+ and χ_p^- being the transverse mode wavefunctions for incident and scattered waves, respectively. For $\mathbf{A} = 0$ in the lead, these functions become identical, implying $k^- = -k^+$. An occupied scattering state (q, k) results in a current $ip(q)$ per unit energy in lead p , comprising an ingoing incident current and an outgoing scattered current. The occupancy of a scattering state (q, k) is dictated by the Fermi function in contact q , allowing the current in terminal p to be expressed in terms of the transmission function T_{pq} , which aggregates transmission probabilities across multiple modes.

Orthogonality of the scattering states: It can be shown that the scattering states together with any bound states do form a complete orthonormal set. Recently the set of scattering states has been used to develop a formal theory not only for the current but also the current fluctuations in coherent conductors (considering the correlations between the wavepackets incident from different leads)

In general the x -direction is different in each of the leads. For example, in Fig. 1.5, x_3 is orthogonal to x_1 and x_2 . We need to choose our gauge such that the vector potential has the form $A \sim \hat{x}_q \mathbf{B} y_q$ in every lead q . It has been shown that this can be done even for conductors with multiple leads arranged arbitrarily. This enables the definition of scattering states in non-zero magnetic fields.

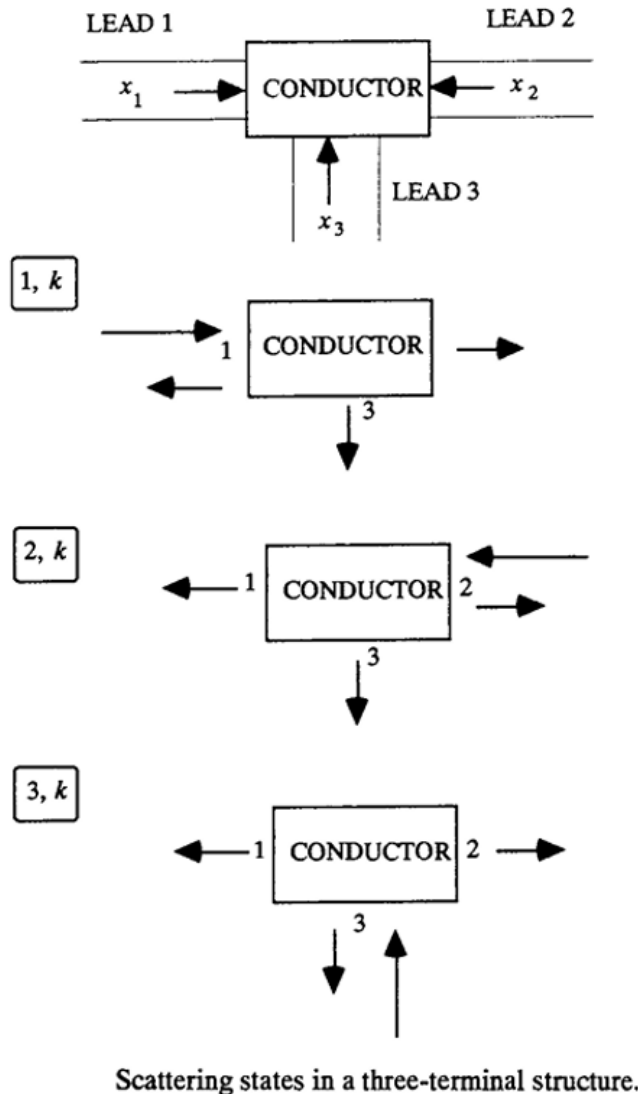


Fig. 1.5

Independent transverse modes: Given any arbitrary potential and field configuration, the transverse modes are not necessarily orthogonal. However, if the orthogonality condition is satisfied, we can obtain the net current by superposing the currents carried by different transverse modes, especially in the presence of a magnetic field (cross-terms vanish).

Non-coherent transport

A conceptual probe with the appropriate properties can be used to simulate the effect of phase-breaking processes. Thus, a partially coherent conductor or an incoherent conductor can be visualized as a fully coherent conductor with a conceptual probe stuck to it. The current in

such a structure can be described by,

$$i_p(E) = \frac{2e}{h} \sum_q \bar{T}_{pq} [f_p(E) - f_q(E)] + \frac{2e}{h} \bar{T}_{p\varphi} [f_p(E) - f_\varphi(E)] \quad (1.19)$$

where the index q runs over all the real terminals while the index φ denotes a fictitious phase-breaking probe attached to the conductor. The current at the fictitious probe is given by,

$$i_q(E) = \frac{2e}{h} \sum_q \frac{2e}{h} \bar{T}_{\varphi q} [f_\varphi(E) - f_q(E)] \quad (1.20)$$

The distribution function at the fictitious probe $f_\varphi(E)$, can be expressed as,

$$f_\varphi(E) = \frac{1}{\bar{R}_\varphi} (i_\varphi + \sum_q \bar{T}_{pq} f_q(E)) \quad \text{where} \quad \bar{R}_\varphi \equiv \sum_q \bar{T}_{\varphi q} \quad (1.21)$$

Thus, we obtain an expression for the current where the phase-breaking probes have been eliminated but their effect is incorporated into a set of effective transmission coefficients among the real terminals:

$$i_p = \frac{2e}{h} \sum_q \frac{2e}{h} \bar{T}_{pq}^{eff} [f_p - f_q] - \frac{2e}{h} [\bar{T}_{p\varphi} / \bar{R}_\varphi] i_\varphi \quad (1.22)$$

where,

$$\bar{T}_{pq}^{eff} = \bar{T}_{pq} + \frac{\bar{T}_{p\varphi} \bar{T}_{\varphi q}}{\bar{R}_\varphi} \quad (1.23)$$

If we neglect the current at the fictitious probe $i_\varphi(E)$, then this has exactly the same form as the current expression for coherent transport without any exclusion principle factors. However, there is no reason to assume that $i_\varphi(E) = 0$. Scattering processes take electrons out from one energy channel and reinject them at another channel. Consequently $i_\varphi(E)$ is negative at some energies (representing outscattering) and positive at some others (representing inscattering). All that we know for sure is that the net current integrated over all energies must be zero,

$$\int i_\varphi(E) dE = 0 \quad (1.24)$$

since the net outscattering must balance the net inscattering.

The function $i_\varphi(E)$ describes the flow of carriers from one energy channel to another, and is referred to as the ‘vertical flow’. In general these vertical currents depend on the Fermi functions in the contacts $f_p(E)$ in a complicated manner due to the exclusion principle. These effects are difficult to accommodate in theory, and the easiest simplification is to neglect vertical currents altogether and set $i_\varphi(E) = 0$.

Non-coherent elastic transport occurs in devices exceeding the phase-relaxation length, where electrons exhibit both vertical and lateral movement. Electrons traverse coherently in one energy channel until they undergo inelastic scattering, transitioning to another energy channel, and this cycle continues. Under these circumstances, the complexity of exclusion principle-related factors complicates current expressions. However, assuming no net vertical flow alleviates these complexities. Electrons scattering from one channel into another have counterparts, balancing the energy exchange and preventing dissipation or energy relaxation.

Influence of vertical flow on resistance: Momentum relaxation drives resistance, while energy relaxation assists in maintaining near-equilibrium energy distributions, often having minimal resistance impact. The non-coherent elastic transport model effectively approximates current flow through a conductor. This description of transport through independent energy channels (undergoing repeated phase-breaking) allows for the exclusion principle factors to be discounted, enabling us to express current via the transmission function similarly to coherent transport despite the non-coherent nature of the channel transitions.

Consider a sample with both elastic and inelastic processes; we desire to compute the current I for a given bias. It may be replaced with a sample containing only elastic scatterers that replicate the same momentum and phase relaxation without energy relaxation—eliminating vertical flow. The current I_{EL} for this elastic sample can be derived using the Landauer-Buttiker formalism. The equivalence of I_{EL} and I depends on the following:

- Uniform transmission: the transmission function $\bar{T}(E)$ remains constant within the transport energy range $\mu_1 + (\sim k_B T) > E > \mu_2 - (\sim k_B T)$; vertical flow does not affect resistance as all energy channels conduct equally. Here, internal transfers caused by vertical flow do not influence net current, enabling accurate results without considering vertical flow ($I_{EL} = I$). However, this applies to net current, not to energy distribution, which is significant for calculating heat dissipation in the conductor. For resistance assessment, neglecting vertical flow may suffice if transmission characteristics remain uniform across energies.
- Non-uniform transmission: the transmission function $\bar{T}(E)$ does not remain constant in that regime. For example, consider a conductor where one side effectively transports energy E_A , and the opposite side does so for E_B , leading to nearly zero transmission probability across the sample. Here, neglecting inelastic scattering leads to isolated channels and no current flow.
- Under such conditions of 'strong localization', vertical flow can substantially enhance the current beyond expectations if inelastic transitions are neglected. However, even in the absence of strong localization, inelastic transitions may still result in a slight current increase, though it might not be experimentally detectable.

1.2 Transmission and Green's Functions

1.2.1 Transmission and S-matrices

A coherent conductor can be characterized at each energy by an S-matrix that relates the outgoing wave amplitudes to the incoming wave amplitudes at the different leads. In general,

$$[b] = [S][a] \quad (1.25)$$

where $[a]$ and $[b]$ are vectors corresponding to incoming and outgoing amplitudes across various modes. The S-matrix elements $s_{m \leftarrow n}$ (corresponding to transport from mode n to mode m) are calculated based on the dynamical equation describing the conductor. Some properties of S-matrix are listed below:

- $\dim(S) = M_T \times M_T$; M_T = total number of modes across the leads. Specifically, $M_T(E) = \sum_p M_p(E)$

- The Transmission probability (from mode n to mode m) is defined as $T_{m \leftarrow n} = |s_{m \leftarrow n}|^2$
- Transmission function (from lead q to lead p) is defined as $T_{p \leftarrow q} = \sum_{m \in q} \sum_{m \in p} T_{m \leftarrow n}$ where the sum is carried out over all modes in corresponding leads.
- Unitarity: $[S]^\dagger [S] = I = [S][S]^\dagger \implies \sum_{m=1}^{M_T} |s_{mn}|^2 = 1 = \sum_{m=1}^{M_T} |s_{nm}|^2$
- Redefinition in terms of current amplitude: $s'_{nm} = \sqrt{v_m/v_n} s_{nm}$ The newly defined matrix $[S']$ is no longer unitary.
- Sum rules: $\sum_q \bar{T}_{qp}(E) = \sum_q \bar{T}_{pq}(E) = M_p(E)$, where $M_p(E)$ is the number of modes in lead p at energy E . The sum rules imply that for two-terminal devices the transmission function is always reciprocal, even if a magnetic field is present ($\bar{T}_{21}(E) = \bar{T}_{12}(E)$).
- For coherent transport, the unitarity of the S-matrix ensures that the sum rule for the conductance matrix is satisfied: $\sum_q G_{pq} = \sum_q G_{qp} = \frac{2e^2}{h} \int M_p(E) \left(-\frac{\partial f_0}{\partial E}\right) dE$
- *Reciprocity: $[S]_{+\mathbf{B}} = [S^T]_{+,-\mathbf{B}} \implies [G_{pq}]_{+\mathbf{B}} = [G_{qp}]_{+,-\mathbf{B}}$

Note that the property of reciprocity holds only for small bias, unlike the sum rules which hold irrespective of the bias. The equality assumes that the electrostatic potential $U(x, y)$ inside the conductor remains unchanged when we reverse the magnetic field. This is clearly not true for large bias since the Hall voltage reverses when the magnetic field reverses. But for an infinitesimally small bias the electrostatic potential inside the conductor is approximately the same as that at equilibrium and it can be shown that $U(x, y)$ is unchanged when the B-field is reversed.

When dealing with a large conductor, it is often convenient to divide it conceptually into several sections whose S-matrices are determined individually. These S-matrices of successive sections can be combined assuming complete coherence, complete incoherence or partial coherence among the sections. This enables the calculation of the transmission function for a partially coherent conductor. Generally, the sample is conceptually divided into little segments whose dimensions are of the order of a phase-relaxation length. The S-matrix for an individual segment is calculated assuming transport to be coherent within the segment. Different segments are then combined incoherently.

The Green's function $G^R(r, r')$ can be viewed as a generalized S-matrix that allows us to describe the response at any point r due to an excitation at point r' . For a single mode in a lead (infinite 1D wire) with constant potential energy and zero vector potential as shown in Fig 1.6, the green's function is given by,

$$G = [E - U_0 + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}]^{-1} \quad (1.26)$$

i.e.,

$$[E - U_0 + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}] G(x, x') = \delta(x - x') \quad (1.27)$$

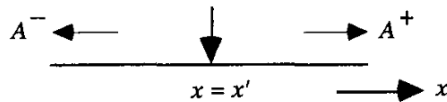


Fig. 1.6

Thus, we can write,

$$G(x, x') = A^+ \exp\{[ik(x - x')]\} \quad x > x'$$

$$G(x, x') = A^- \exp\{[-ik(x - x')]\} \quad x < x'$$

where $k = [2m(E - U_0)]^{1/2}/\hbar$. Using continuity of G and appropriate discontinuity of $\frac{\partial G(x, x')}{\partial x}$ at $x = x'$ we get,

$$A^+ = A^- \quad \text{and} \quad ik[A^+ + A^-] = \frac{2m}{\hbar^2} \quad (1.28)$$

Hence,

$$A^+ = A^- = -\frac{i}{\hbar v} \quad ; \quad v \equiv \frac{\hbar k}{m} \quad (1.29)$$

and the retarded Green's function (describing outgoing waves that originate at the point of excitation) is given by,

$$G^R(x, x') = -\frac{i}{\hbar v} \exp\{[ik|x - x'|]\} \quad (1.30)$$

The corresponding advanced Green's function (describing incoming waves that disappear at the point of excitation) is given by,

$$G^A(x, x') = \frac{i}{\hbar v} \exp\{[-ik|x - x'|]\} \quad (1.31)$$

Alternatively, we can write,

$$G^R = [E + i\eta - H_{op}]^{-1} \quad \eta \rightarrow 0^+$$

$$G^A = [E - i\eta - H_{op}]^{-1} \quad \eta \rightarrow 0^+$$

For a multi-moded wire, the retarded Green's function is given by,

$$G^R(x, x') = \sum_m A_m^\pm \chi_m(y) \exp\{[ik_m|x - x'|]\} \quad (1.32)$$

where $A_m^\pm$ are the amplitudes of the different modes that propagate away from the source. The transverse mode wavefunctions, $\chi_m(y)$ satisfy (with $B = 0$),

$$[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} + U(y)]\chi_m(y) = \varepsilon_{m,0} \chi_m(y) \quad (1.33)$$

where $U(y)$ is the transverse confining potential in the y -direction. For given $U(x, y)$ and B configuration these functions are orthogonal,

$$\int \chi_n(y) \chi_m(y) dy = \delta_{nm} \quad (1.34)$$

since they satisfy the same equation with different eigenvalues. However, the transverse modes are not necessarily orthogonal for an arbitrary $U(x, y)$ and B configuration. We will assume that these functions $\chi_m(y)$ are real.

As before, we utilize the continuity and discontinuity at $x = x'$ to get,

$$G^R(x, y; x', y') = \sum_m -\frac{i}{\hbar v_m} \chi_m(y) \chi_m(y') \exp\{[ik_m|x - x'|]\} \quad (1.35)$$

where $k_m = [2m(E - \varepsilon_{m,0})]^{1/2}/\hbar$.

Since the eigenfunctions ($H_{op}\psi_\alpha(\mathbf{r}) = \varepsilon_\alpha\psi_\alpha(\mathbf{r})$) of the hamiltonian form a complete basis set, we can write,

$$G^R(\mathbf{r}, \mathbf{r}') = \sum_\alpha \frac{\psi_\alpha(\mathbf{r})\psi_\alpha^*(\mathbf{r}')}{E - \varepsilon_\alpha + i\eta}$$

$$G^A(\mathbf{r}, \mathbf{r}') = \sum_\alpha \frac{\psi_\alpha(\mathbf{r})\psi_\alpha^*(\mathbf{r}')}{E - \varepsilon_\alpha - i\eta}$$

S-Matrix and Green's Functions

Consider a conductor connected to contacts via two leads p and q as shown in Fig 1.7.

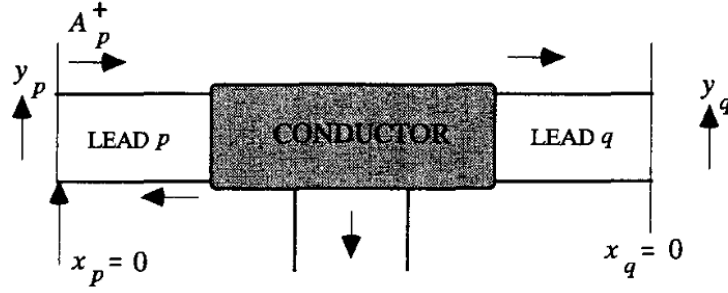


Fig. 1.7

The Green's function between a point lying on the plane $x_p = 0$ and another point lying on $x_q = 0$ can be expressed in terms of the relevant S-matrix elements as,

$$G_{qp}^R = \delta_{qp} A_p^- + s'_{qp} A_p^+ \quad (1.36)$$

where A_p^+ and A_p^- are the wave amplitudes for wave going toward and away from the conductor respectively. s'_{qp} describes the transmission from lead p to lead q . using A_p^+ and A_p^- we can write,

$$s_{qp} = -\delta_{qp} + i\hbar\sqrt{v_m v_n} G_{qp}^R \quad (1.37)$$

This is known as the Fisher-Lee relation describing S-matrix elements in terms of relevant Green's functions.

For a multi-moded wire the green's function (across leads p and q) can be written in terms of S-matrix elements as,

$$G_{qp}^R(y_q; y_p) = \sum_{m \in q} \sum_{n \in p} \frac{-i}{\hbar \sqrt{v_n v_m}} \chi_n(y_q) [\delta_{nm} + s_{nm}] \chi_m(y_p) \quad (1.38)$$

Using orthogonality of transverse mode wavefunctions $\chi_m(y)$ the resultant Fisher-Lee rela-

tion is given by,

$$s_{nm} = -\delta_{nm} + i\hbar\sqrt{\nu_n\nu_m} \int \int \chi_n(y_q) [G_{qp}^R(y_q; y_p)] \chi_m(y_p) dy_q dy_p \quad (1.39)$$

Discretization: the method of finite differences

For an arbitrarily shaped conductor, we have,

$$[E + i\eta - H_{op}(\mathbf{r})]G^R(\mathbf{r}; \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}') \quad (1.40)$$

where,

$$H_{op}(\mathbf{r}) = \frac{(i\hbar\nabla + e\mathbf{A})^2}{2m} + U(\mathbf{r}) \quad (1.41)$$

A common approach for solution involves discretization of the spatial coordinate so that the Green's function becomes a matrix: $\mathbf{G}^R(\mathbf{r}, \mathbf{r}') \rightarrow \mathbf{G}^R(\mathbf{i}, \mathbf{j})$ where (i, j) denote points on a discrete lattice. The differential equation becomes a matrix equation,

$$[(E + i\eta)I - H]G^R = I \quad (1.42)$$

where $[I]$ is the identity matrix and $[H]$ is the matrix representation of the Hamiltonian operator H_{op} .

Consider the Hamiltonian with zero vector potential in 1D, $H_{op} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + U(x)$. The action of H_{op} on any function F on discrete lattice ($x = ja$; $j \in \mathbb{Z}$) is given by,

$$[H_{op}F]_{x=ja} = [-\frac{\hbar^2}{2m} \frac{d^2 F}{dx^2}]_{x=ja} + U_j F_j \quad (1.43)$$

where $U_j = U(x = ja)$, $F_j = F(x = ja)$. Under the assumption of small a , using the finite difference method we can write,

$$[\frac{dF}{dx}]_{x=(j+1/2)a} \rightarrow \frac{1}{a}[F_{j+1} - F_j] \quad (1.44)$$

and,

$$[\frac{d^2 F}{dx^2}]_{x=ja} \rightarrow \frac{1}{a^2}[F_{j+1} - 2F_j + F_{j-1}] \quad (1.45)$$

Thus, we can write,

$$[H_{op}F]_{x=ja} = (U_j + 2t)F_j - t(F_{j+1} + F_{j-1}) \quad ; \quad t = \hbar^2/2ma^2 \quad (1.46)$$

or,

$$[H_{op}F]_{x=ja} = \sum_i H(j, i)F_i \quad (1.47)$$

where,

$$H(j, i) = \begin{cases} U_i + 2t & \text{if } i = j, \\ -t & \text{if } i \text{ and } j \text{ are nearest neighbors,} \\ 0 & \text{otherwise.} \end{cases} \quad (1.48)$$

In 2D, this is extended as follows:

$$[H]_{ij} = \begin{cases} U(r_i) + zt & \text{if } i = j, \\ -\tilde{t}_{ij} & \text{if } i \text{ and } j \text{ are nearest neighbors,} \\ 0 & \text{otherwise.} \end{cases} \quad (1.49)$$

where,

$$\tilde{t}_{ij} = t \exp\{[ie\mathbf{A} \cdot (\mathbf{r}_i - \mathbf{r}_j)/\hbar]\} \quad (1.50)$$

Truncating the matrix

We need to invert:

$$G^R = [(E + i\eta)I - H]^{-1} \quad (1.51)$$

Consider a conductor connected to the lead as shown in Fig 1.8:

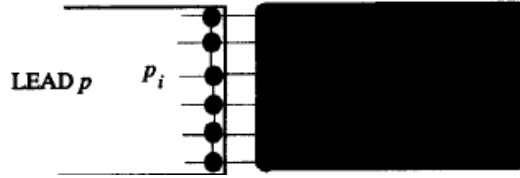


Fig. 1.8

The matrix is infinite dimensional for a finite size conductor connected to semi-infinte leads. We can partition the overall Green's function in Eq.(3.5.10) into submatrices as follows:

$$\begin{bmatrix} G_p & G_{pC} \\ G_{Cp} & G_C \end{bmatrix} = \begin{bmatrix} (E + i\eta)I - H_p & \tau_p \\ \tau_p^\dagger & EI - H_C \end{bmatrix}^{-1} \quad (1.52)$$

where the matrix $[(E + i\eta)I - H_{op}]$ represents the isolated lead, while the matrix $[EI - H_C]$ represents the isolated conductor. We could add an infinitesimal imaginary term $i\eta$ to the conductor matrix as well but it is not necessary, the coupling to the leads effectively gives rise to a finite imaginary term that will swamp it. The coupling matrix is non-zero only for adjacent points i and p_i $\tau_p(p_i, i) = t$ such that,

$$G_C = [EI - H_C - \tau_p^\dagger g_p^R \tau_p]^{-1} \quad (1.53)$$

with, $\tau_p^\dagger g_p^R \tau_p = t^2 g_p^R(p_i, p_j)$. Note that the matrices in this expression are finite matrices of dimensions $(C \times C)$, C being the number of points inside the conductor. But the infinite lead is taken into account **exactly** through the term $\tau_p^\dagger g_p^R \tau_p$.

Assuming that different leads are independent so that their effects are additive the total isolated Green's function for the conductor is given by (using G^R instead of G_C),

$$G^R = [EI - H_C - \Sigma^R]^{-1} \quad (1.54)$$

where

$$\Sigma^R = \sum_p \Sigma_p^R \quad \text{and} \quad \Sigma_p^R(i, j) = t^2 g_p^R(p_i, p_j) \quad (1.55)$$

is called the self energy due to leads, encoding the interaction of the conductor with the leads. For a semi-infinite discrete wire it can be shown that the Green's function between two points along the edge is given by,

$$g^R(p_i, p_j) = -\frac{1}{t} \sum_m \chi_m(p_i) \exp\{[ik_m a]\} \chi_m(p_j) \quad (1.56)$$

Thus, we get,

$$\Sigma_p^R(i, j) = -t \sum_{m \in p} \chi_m(p_i) \exp\{[ik_m a]\} \chi_m(p_j) \quad (1.57)$$

Note that it is not necessary to assume the leads to have regular shapes with well-defined transverse modes, though we have done so for convenience. Even for irregularly shaped leads we can in principle calculate the Green's function and obtain the appropriate self-energy.

Originally, for isolated system we had,

$$H_C \psi_\alpha = \varepsilon_{\alpha 0} \psi_\alpha \quad (1.58)$$

For an open system coupled to leads, we can write,

$$[H_C + \Sigma^R] \psi_\alpha = \varepsilon_\alpha \psi_\alpha \quad (1.59)$$

where the new eigenenergies are given by,

$$\varepsilon_\alpha = \varepsilon_{\alpha 0} - \Delta_\alpha - i(\gamma_\alpha/2) \quad (1.60)$$

such that the time dependence of the eigenstate is modified as,

$$\exp\{[-i\varepsilon_\alpha t/\hbar]\} \rightarrow \exp\{[-i(\alpha 0 - \Delta_\alpha)t/\hbar]\} \exp\{[-\gamma_\alpha t/\hbar]\}$$

Δ_α represents the shift in the energy due to the modification of the dynamics of an electron inside the conductor by the interaction with the leads. On the other hand, the imaginary part of the energy γ_α reflects the fact that an electron injected anywhere in the conductor will eventually disappear through one of the leads stretching out to infinity. Taking the squared magnitude of the wavefunction we obtain the probability:

$$|\psi_\alpha|^2 \propto \exp[-2\gamma_\alpha t/\hbar]$$

The quantity $\hbar/2\gamma_\alpha t$ represents the 'lifetime' or the average time an electron remains in state α before it escapes out into the leads. The lifetime is infinite if the self-energy is Hermitian ($\Gamma = 0$).

We can now define the Transmission function in terms of Green's functions (using Fisher-Lee relation) as follows:

$$\bar{T}_{pq} = Tr_s[\Gamma_p G^R \Gamma_q G^A] \quad (1.61)$$

where,

$$[\Gamma_p]_{ij} = \sum_m \chi_m(p_i) \frac{\hbar v_m}{a} \chi_m(p_j) \quad (1.62)$$

Alternatively,

$$\Gamma_p = i[\Sigma_p^R - \Sigma_p^A] \quad (1.63)$$

and,

$$\Gamma = \sum_p \Gamma_p = \sum_p i[\Sigma_p^R - \Sigma_p^A] = i[\Sigma^R - \Sigma^A] \quad (1.64)$$

Also, we can define the spectral function as,

$$A \equiv i[G^R - G^A] = G^R \Gamma G^A = G^A \Gamma G^R \quad (1.65)$$

Alternatively,

$$A(\mathbf{r}, \mathbf{r}') = \sum_{\alpha} \psi_{\alpha}(\mathbf{r}) \phi_{\alpha}^*(\mathbf{r}') \frac{\gamma_{\alpha}}{(E - \varepsilon_{\alpha 0} + \Delta_{\alpha})^2 + (\gamma_{\alpha}/2)^2} \quad (1.66)$$

where $[H_C + \Sigma^R]\psi_{\alpha} = \varepsilon_{\alpha}\psi_{\alpha}$ and $[H_C + \Sigma^A]\phi_{\alpha} = \varepsilon_{\alpha}^*\phi_{\alpha}$.

As the coupling to the leads is increased, peaks (i) **shift by Δ_{α}** (ii) **broaden by γ_{α}** . When the coupling is very large the peaks all merge and the peaks corresponding to different eigenstates can no longer be distinguished. Furthermore, since the self-energy itself is a function of E, consequently the eigenfunctions ϕ_{α} , ψ_{α} as well as the eigenvalues ε_{α} are functions of E and the peaks in the spectral function can have very non-Lorentzian shapes.

The trace of the spectral function represents the density of states:

$$N[E] = \frac{1}{2\pi} \text{Tr}[A(E)] \quad (1.67)$$

where,

$$\text{Tr}[A(E)] = \int A(\mathbf{r}, \mathbf{r}') d\mathbf{r} = \sum_{\alpha} \frac{\gamma_{\alpha}}{(E - \varepsilon_{\alpha 0} + \Delta_{\alpha})^2 + (\gamma_{\alpha}/2)^2}$$

Thus, we can write,

$$N(E) \rightarrow \sum_{\alpha} \delta(E - \varepsilon_{\alpha 0} + \Delta_{\alpha}) \quad \text{as} \quad \gamma_{\alpha} \rightarrow 0 \quad (1.68)$$

which is in agreement with our expectation for long-lived eigenstates with well-defined energies.

We can also define the local density of states (LDOS) as,

$$\rho(\mathbf{r}, E) = \frac{1}{2\pi} A(\mathbf{r}, \mathbf{r}', E) = -\frac{1}{\pi} \text{Im}[G^R(\mathbf{r}, \mathbf{r}'; E)] \quad (1.69)$$

Thus, we have,

$$\begin{aligned} \rho(\mathbf{r}, E) &\sim \sum_{\alpha} \frac{1}{2\pi} \frac{\gamma_{\alpha}}{(E - \varepsilon_{\alpha 0} + \Delta_{\alpha})^2 + (\gamma_{\alpha}/2)^2} \psi_{\alpha}(\mathbf{r}) \phi_{\alpha}^*(\mathbf{r}') \\ &\rightarrow 0 \quad \text{as} \quad \gamma_{\alpha} \rightarrow 0 \end{aligned}$$

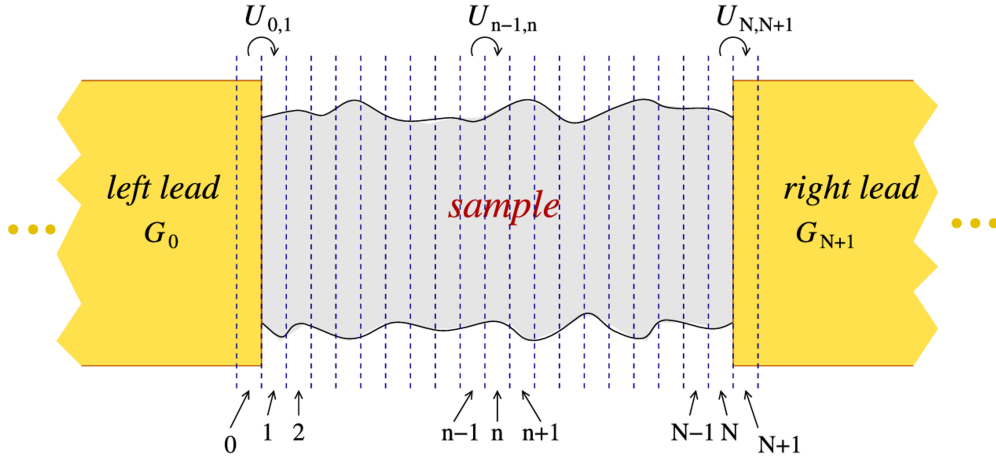
again in agreement with the expectation for long-lived eigenstates with well-defined energies.

Chapter 2

Recursive Green's Functions: The Method

2.1 Scheme

The two-probe setup including a sample (conductor) is shown in Fig 2.1.



Typical two-probe RGF scheme used in the numerical calculations

Fig. 2.1

The conductor is divided into N slices, and each slice can have a maximum of M sites. The 0^{th} and the $N+1^{th}$ slice refer to the slices of the leads that are in contact with the conductor on the left and right side respectively. The goal is to write the Green's function for the entire system. We derive the Green's functions by recursively determining the Green's functions for each slice. The steps are described below.

2.1.1 Left Green's functions

We start with the Green's function of the left lead. Let this be g_L . We denote the Hamiltonian of the isolated n^{th} slice by h_n and its corresponding Green's function by g_n . For the left sweep, n goes from 0 to N . We need to calculate the matrix elements $G_{0,0}^L$ and $G_{0,1}^L$. For this, we take

the unperturbed Green's function as,

$$G^{(0)} = |0\rangle g_l \langle 0| + |1\rangle g_1 \langle 1| \quad (2.1)$$

Here and in later sections, $|n\rangle$ represents the electronic wavefunction in the n^{th} slice. Next, we define the connection matrix as,

$$V = |0\rangle U_{0,1} \langle 1| + |1\rangle U_{1,0} \langle 0| \quad (2.2)$$

Here, the matrix elements $U_{0,1}$ and $U_{1,0}$ connect the left lead to the $n=1$ slice. Using Dyson's equation, we write the elements of the left Green's function as,

$$\begin{aligned} G_{1,1}^L &= \langle 1| G^{(0)} + G^{(0)} V G |1\rangle \\ &= \langle 1| G^{(0)} |1\rangle + \sum_{m,m'} \langle 1| G^{(0)} |m\rangle \langle m| V |m'\rangle \langle m'| G^L |1\rangle \\ &= g_1 + g_1 U_{1,0} G_{0,1}^L \end{aligned} \quad (2.3)$$

and

$$\begin{aligned} G_{0,1}^L &= \langle 0| G^{(0)} + G^{(0)} V G_L |1\rangle \\ &= \langle 0| G^{(0)} |1\rangle + \sum_{m,m'} \langle 0| G^{(0)} |m\rangle \langle m| V |m'\rangle \langle m'| G^L |1\rangle \\ &= g_L U_{0,1} G_{1,0}^L \end{aligned} \quad (2.4)$$

Using equation 2.3 and 2.4, we can write,

$$\begin{aligned} G_{1,1}^L &= (I - g_1 U_{1,0} g_L U_{0,1})^{-1} g_1 \\ &= (E - h_1 - U_{1,0} g_L U_{0,1})^{-1} \end{aligned} \quad (2.5)$$

Here, we have used that $g_1 = (E - h_1)^{-1}$. The rest of the matrix elements of the left Green's function are written in the following way. We write the Dyson's equation again by considering the unperturbed Green's function and the perturbation to be,

$$\begin{aligned} G_0 &= \sum_{i,i'=1}^{n-1} |i\rangle G^L \langle i'| + |n\rangle g_n \langle n| \\ V &= |n-1\rangle U_{n-1,n} \langle n| + |n\rangle U_{n,n-1} \langle n-1| \end{aligned} \quad (2.6)$$

The Dyson's equation reads,

$$\begin{aligned} G_{n,n}^L &= \langle n| G^{(0)} |n\rangle + \sum_{i,i'=1} \langle n| G^{(0)} V G^L |n\rangle \\ &= g_n + g_n U_{n,n-1} G_{n-1,n}^L \end{aligned} \quad (2.7)$$

$$\begin{aligned} G_{n-1,n}^L &= \langle n-1| G^{(0)} |n\rangle + \sum_{i,i'=1} \langle n-1| G^{(0)} V G^L |n\rangle \\ &= G_{n-1,n-1}^L U_{n-1,n} G_{n,n}^L \end{aligned} \quad (2.8)$$

Using the equations, we write,

$$G_{n,n}^L = g_n + g_n U_{n,n-1} G_{n-1,n-1}^L U_{n-1,n} G_{n,n}^L \quad (2.9)$$

or,

$$G_{n,n}^L = (E - h_n - U_{n,n-1} G_{n-1,n-1}^L U_{n-1,n})^{-1} \quad (2.10)$$

Also,

$$G_{0,n}^L = G_{0,n-1}^L U_{n-1,n} G_{n,n}^L \quad (2.11)$$

2.1.2 Right Green's function

We determine the relevant matrix elements of the Right Green's functions in the same way as we did for the left Green's function. The matrix elements are given below,

$$G_{N,N}^R = (E - h_N - U_{N,N+1} g_R U_{N+1,N})^{-1} \quad (2.12)$$

$$G_{n,n}^R = (E - h^n - U_{n,n+1} G_{n+1,n+1}^R U_{n+1,n})^{-1} \quad (2.13)$$

$$G_{N+1,n}^R = G_{N+1,n+1}^R U_{n+1,n} G_{n,n}^R \quad (2.14)$$

Here, n goes from 1 to N+1. The slicing scheme is demonstrated in Fig 2.2.

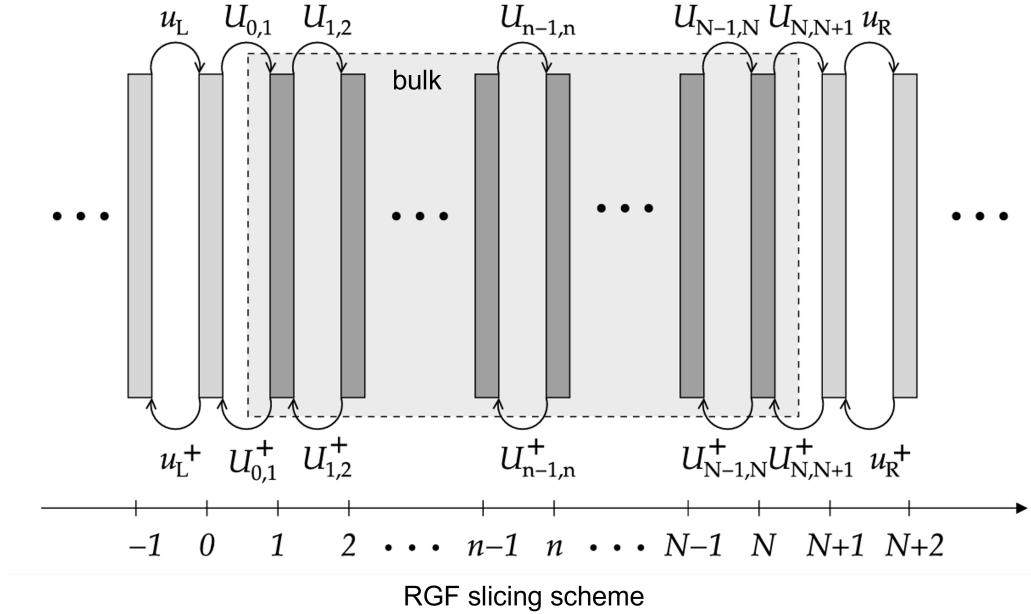


Fig. 2.2

2.1.3 Full Green's function

The full Green's function can be written from the matrix elements of the left and the right Green's function, We take the unperturbed Green's function and the perturbation to be,

$$\begin{aligned} G^{(0)} &= g_n + G_{n-1,n-1}^L + G_{n+1,n+1}^R \\ V &= U_{n-1,n} + U_{n,n-1} + U_{n,n+1} + U_{n+1,n} \end{aligned} \quad (2.15)$$

Applying the above relations in the Dyson equation, we find,

$$\begin{aligned} G_{n,n} &= g_n + \sum_{m',m} \langle n | G^{(0)} | m \rangle \langle m | V | m' \rangle \langle m' | G | n \rangle \\ &= g_n + g_n (U_{n,n-1} G_{n-1,n} + U_{n,n+1} G_{n+1,n}) \end{aligned} \quad (2.16)$$

$$G_{n-1,n} = G_{n-1,n-1}^L U_{n-1,n} G_{n,n} \quad (2.17)$$

$$G_{n+1,n} = G_{n+1,n+1}^R U_{n+1,n} G_{n,n} \quad (2.18)$$

Substituting for the connection matrix elements, we get

$$\boxed{G_{n,n} = [I - g_n (U_{n,n-1} G_{n-1,n-1}^L U_{n-1,n} + U_{n,n+1} G_{n+1,n+1}^R U_{n+1,n})]^{-1}} \quad (2.19)$$

Besides this, we also have,

$$\boxed{G_{0,n} = G_{0,n-1}^L U_{n-1,n} G_{n,n}} \quad (2.20)$$

$$\boxed{G_{N+1,n} = G_{N+1,n+1}^R U_{n+1,n} G_{n,n}} \quad (2.21)$$

The quantity $G_{n,n}$ is required to calculate the local density of states. However, if we want to calculate the transmission matrices of the entire setup, we only need the matrix elements $G_{0,0}$, $G_{0,N+1}$, $G_{N+1,0}$, $G_{N+1,N+1}$. This can be done without determining the matrix element of the Green's function corresponding to each slice (i.e., $G_{n,n}$, $G_{0,n}$, $G_{n,0}$). To do this, we close the left sweep with the right lead and the right lead with the left lead.

- For the left sweep,

$$\begin{aligned} G_{0,N+1} &= G_{0,N}^L U_{N,N+1} G_{N+1,N+1} \\ G_{N+1,N+1} &= (E - h_R - U_{N+1,N} G_{N,N}^L U_{N,N+1})^{-1} \end{aligned} \quad (2.22)$$

- For the right sweep,

$$\begin{aligned} G_{N+1,0} &= G_{N,1}^R U_{1,0} G_{0,0} \\ G_{0,0} &= (E - h_L - U_{0,1} G_{1,1}^R U_{1,0})^{-1} \end{aligned} \quad (2.23)$$

2.1.4 Lead's Green function

To calculate the matrix elements recursively, we have taken Green's function of the leads as inputs, along with the connection matrices. The leads in this setup are semi-infinite leads, and while computing their Green's function, we need to take care of two things:

- The leads are metallic with a high density of states: We tune the chemical potential to maximize the DOS.

- The leads form ohmic contacts with the conductor at the interface, thus reducing the contact resistance - We model the leads accordingly to reduce the contact resistance due to band structure mismatch.

In the paper, we consider three models for the lead - square lattice, Honeycomb lattice with armchair edges and Honeycomb lattice with zigzag edges. They solve for Green's function analytically for square lattice, and then they provide two methods that can be used to compute the Green's function numerically - Eigenmode decomposition method, and the decimation approach.

Analytic approach

We use tight-binding square lattices in the lead. Since the lead is semi-infinite in x -direction and finite in the y -direction, electron wavefunction is extended in the x -direction and quantized in the y -direction. The transverse wavefunction can then be written as,

$$\chi_\nu(j) = \sqrt{\frac{2}{M+1}} \sin\left(\frac{\pi\nu j}{M+1}\right) \quad (2.24)$$

Here, $j = 0$ and $j = M+1$ are the hard wall boundaries in the transverse direction, and $\nu = 1, \dots, M$.

In the longitudinal direction, there is hard wall boundary condition at $x = N-1$. The longitudinal wavefunction can then be written as,

$$\phi_\mu(n) = \sqrt{\frac{2}{\pi}} \sin[\mu(n - N + 1)] \quad (2.25)$$

Here, μ is the longitudinal quantum number. Comparing this with the continuum model, we can identify $\mu = k_x^\mu a_x$, k_x^μ being the wave vector. The dispersion relation for this model is

$$E_{\nu\mu} = V_{lead} - 2t_x \cos\mu - 2t_y \cos\left(\frac{\pi\nu}{M+1}\right) \quad (2.26)$$

The Green's function can then be written as,

$$g(n, j, n', j'; E) = \int_0^\pi d\mu \sum_{\nu=1}^M \frac{[\phi_\mu(n)\chi_\nu(j)] * [\phi_\mu(n')\chi_\nu(j')]}{E - E_{\nu\mu} + i0^+} \quad (2.27)$$

Since only the surface of the lead is relevant here, we set $n = n' = N$. Also, we set $E=0$. The Green's function then becomes,

$$g_L(j, j') = \frac{2}{\pi} \sum_{\nu=1}^M \chi_\nu(j) * \chi_\nu(j') \int_0^\pi d\mu \frac{\sin^2\mu}{p + q \cos\mu} \quad (2.28)$$

Here, $p \equiv V_{lead} + 2t_y \cos\left(\frac{\pi\nu}{M+1}\right) + i)^+$, and $q \equiv 2t_x$

Integrating over μ , we can write,

$$G_N^{semi}(j, j') = \sum_{\nu=1}^M \chi_\nu * (j) \tilde{G}^{semi}(\nu) \chi_\nu(j') \quad (2.29)$$

$$\tilde{G}^{semi}(\nu) = \frac{2p}{q^2} \left[1 - \sqrt{1 - \left(\frac{q}{p}\right)} \right]$$

This approach can be used to model honeycomb lattices with armchair edges as well. However, we cannot use this approach for zigzag edges due to the mixing of modes.

Eigenmode decomposition

In this approach, we use the fact that the semi-infinite leads will remain the same if another slice of lattice is added to the transverse edge. Thus, we can write (from eq.(10)),

$$g_L = (E - h_1 - t_x^2 g_L)^{-1} \quad (2.30)$$

Here, we have used that, $U_{0,1} = U_{1,0} = t_x^2 I$ for square lattices. Since h_1 is hermitian, we can diagonalize the equation 30 using the unitary transformation,

$$\begin{aligned} T^\dagger h_1 T &= \text{diag}(\epsilon_\nu) \\ T^\dagger g_L T &= \text{diag}(\tilde{g}_L(\nu)) \end{aligned} \quad (2.31)$$

Hence, we have

$$\tilde{g}_L(\nu) = (E - \epsilon_\nu - t_x^2 \tilde{g}_L(\nu))^{-1} \quad (2.32)$$

Solving for $\tilde{g}_L(\nu)$, we have,

$$\tilde{g}_L(\nu) = \frac{(E - \epsilon_\nu)}{2t_x^2} \pm \sqrt{\frac{(E - \epsilon_\nu)^2}{4t_x^4} - \frac{1}{t_x^2}} \quad (2.33)$$

When $|E - \epsilon_\nu| < 2t_x$, $\tilde{g}_L^r(\nu)$ has a finite imaginary part. However, when $|E - \epsilon_\nu| \geq 2t_x$ we need to put a finite imaginary part by hand, so that the Green's function of the sample has a finite imaginary part. Thus, we have

$$\tilde{g}_L^r(\nu) = \begin{cases} \frac{(E - \epsilon_\nu)}{2t_x^2} \left[1 - \sqrt{1 - \frac{4t_x^2}{(E - \epsilon_\nu)^2}} \right] + i0^+, & |E - \epsilon_\nu| \geq 2t_x, \\ \frac{(E - \epsilon_\nu)}{2t_x^2} - i\sqrt{\frac{1}{t_x^2} - \frac{(E - \epsilon_\nu)^2}{4t_x^4}}, & |E - \epsilon_\nu| < 2t_x. \end{cases} \quad (2.34)$$

For square lattice, the unitary matrix used for transformation consists of the transverse wavefunctions ($\chi_\nu(j)$). Hence,

$$\begin{aligned} T_{\nu j} &= \sqrt{\frac{2}{M+1}} \sin\left(\frac{\pi \nu j}{M+1}\right) \\ \epsilon_\nu &= V_{lead} - 2t_y \cos\left(\frac{\pi \nu}{M+1}\right); j, \nu = 1, \dots, M \end{aligned} \quad (2.35)$$

The Green's function is,

$$g_L(j, j') = \sum_{\nu=1}^M \chi_{\nu}(j) \tilde{g}_L(\nu) \chi_{\nu}(j') \quad (2.36)$$

This method cannot be used for a honeycomb lattice since $U_{0,1}$ is not proportional to identity, and it does not commute with the slice Hamiltonian.

Decimation method

In this method, we divide the lead into slices. Each slice is then modeled as a site with an arbitrary basis of dimension J , and we consider only nearest neighbor hoppings between the sites. The equation,

$$(E - H)G(E) = I \quad (2.37)$$

is then solved using this basis. The connection between sites is represented by u (left to right) and $u^\dagger$ (right to left). This is demonstrated in Fig. 2.3.

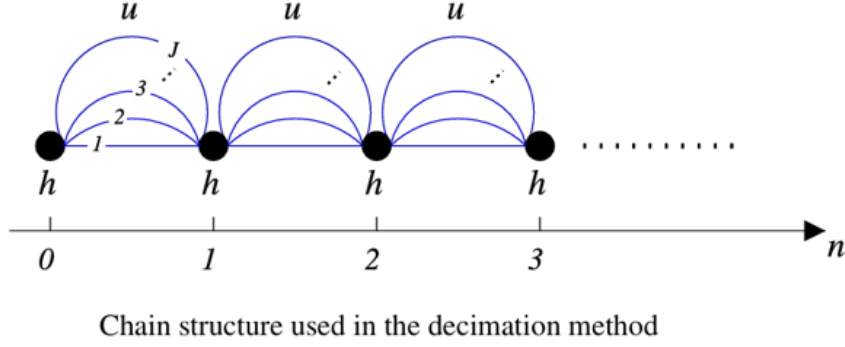


Fig. 2.3

Expanding equation Eq.2.37 in the mentioned basis, we get

$$\sum_{n,l,m} \langle n | (E - H) | m \rangle \langle m | G(E) | l \rangle \quad (2.38)$$

$$(E-H) = \begin{bmatrix} E-h & -u & 0 & 0 & \dots & 0 \\ -u^\dagger & E-h & -u & 0 & \dots & 0 \\ \vdots & \ddots & & & & \\ 0 & \dots & 0 & 0 & -u^\dagger & E-h \end{bmatrix} \mathbf{G} = \begin{bmatrix} G_{0,0} & G_{0,1} & \dots \\ G_{1,0} & G_{1,1} & \dots \\ \vdots & \ddots & \\ G_{n,0} & G_{n,1} & \dots \\ \vdots & \dots & \end{bmatrix}$$

We consider the product of all the rows of the matrix $(E - H)$ with the first column of the matrix $\mathbf{G}$. The relevant equations are,

$$\begin{aligned} (E-h)G_{0,0} &= I + uG_{1,0} \\ (E-h)G_{n,0} &= uG_{n+1,0} + u^\dagger G_{n-1,0}, \quad \forall n > 0 \end{aligned} \quad (2.39)$$

Using equation 2.39, we can write,

$$(E - h)G_{0,0} = I + u(E - h)^{-1}(uG_{0,0} + u^\dagger G_{0,0}) \quad (2.40)$$

or,

$$[E - h - u(E - h)^{-1}u^\dagger]G_{0,0} = I + u(E - h)^{-1}uG_{2,0} \quad (2.41)$$

Again, using equation 2.39, we write,

$$\begin{aligned} G_{n+1,0} &= (E - h^{-1}(uG_{n+2,0} + u^\dagger G_{n,0})) \\ G_{n-1,0} &= (E - h^{-1}(u^\dagger G_{n-2,0} + uG_{n,0})) \end{aligned} \quad (2.42)$$

Together with equation 2.39, we can write,

$$(E - h)G_{n,0} = u(E - h)^{-1}(uG_{n+2,0} + u^\dagger G_{n,0}) + u^\dagger(E - h)^{-1}(uG_{n,0} + u^\dagger G_{n-2,0}) \quad (2.43)$$

or,

$$[E - h - u(E - h)^{-1}u^\dagger - u^\dagger(E - h)^{-1}u]G_{n,0} = u(E - h)^{-1}uG_{n+2,0} + u^\dagger(E - h)^{-1}u^\dagger G_{n-2,0} \quad (2.44)$$

Equations 2.41 and 2.44 generate a new recursion that involves only even sites.

$$\begin{aligned} (E - \epsilon_1^s)G_{0,0} &= I + \alpha_1 G_{2,0} \\ (E - \epsilon_1)G_{n,0} &= \alpha_1 G_{n+2,0} + \beta_1 G_{n-2,0}; \quad \forall n > 0 \end{aligned} \quad (2.45)$$

with,

$$\begin{aligned} \alpha_1 &= u(E - h)^{-1}u \\ \beta_1 &= u^\dagger(E - h)^{-1}u^\dagger \\ \epsilon_1^s &= h + u(E - h)^{-1}u^\dagger \\ \epsilon_1 &= \epsilon_1^s + u^\dagger(E - h)^{-1}u \end{aligned} \quad (2.46)$$

Repeating the procedure that we did to get this new recursion formula with the even sites k times, we can then relate the Green's functions of the sites that are multiples of 2^k .

$$\begin{aligned} (E - \epsilon_k^s)G_{0,0} &= I + \alpha_k G_{2,0} \\ (E - \epsilon_k)G_{2^k n,0} &= \alpha_k G_{2^k(n+2),0} + \beta_k G_{2^k(n-2),0}; \quad \forall n > 0 \end{aligned} \quad (2.47)$$

with,

$$\begin{aligned} \alpha_k &= \alpha_{k-1}(E - h)^{-1}\alpha_{k-1} \\ \beta_k &= \beta_{k-1}(E - h)^{-1}\beta_{k-1} \\ \epsilon_k^s &= \epsilon_{k-1} + \alpha_{k-1}(E - h)^{-1}\beta_{k-1} \\ \epsilon_k &= \epsilon_k^s + \beta_{k-1}(E - h)^{-1}\alpha_{k-1} \end{aligned} \quad (2.48)$$

The decimation process stops when $||\alpha_k||$, and $||\beta_k||$ are sufficiently small. In this case, we approximate,

$$G_{0,0} \approx (E - \epsilon_k^s)^{-1} \quad (2.49)$$

This is the surface Green's function that can be used as the input function for lead while

calculating the Green's function of the system. The finite imaginary part has to be added by hand while deriving the Green's function by this method.

This concludes the computation of necessary Green's functions using recursive Green's function (RGF) method. In the following section we demonstrate the use of RGF method for graphene to calculate some important physical quantities.

Chapter 3

Applications

3.1 Target: Graphene

3.1.1 Structure, model

Graphene is a two-dimensional material consisting of a single layer of carbon atoms arranged in a hexagonal (honeycomb) lattice as shown in Fig 3.1. The electronic structure of graphene is gapless because of its linear dispersion relation near the Dirac points.

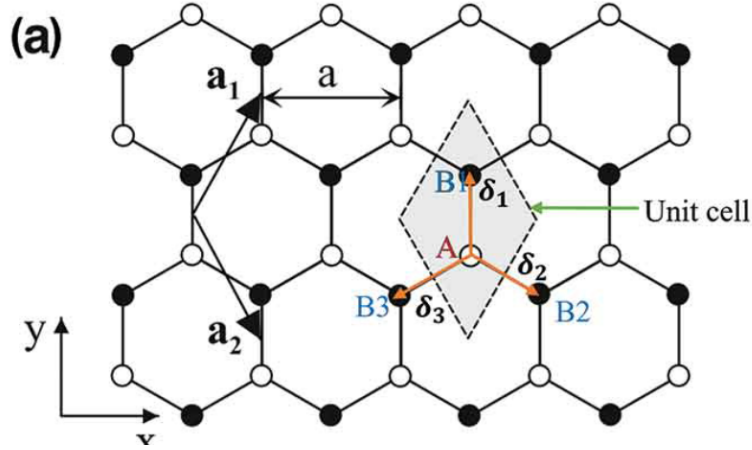


Fig. 3.1

The honeycomb lattice in graphene can be described using two inequivalent sublattices, often labeled as A (formed by join black dots) and B (formed by join white dots) that govern its electronic properties. Each carbon atom in one sublattice is connected to three nearest neighbors in the other sublattice. In this case the electronic behavior near the Fermi level can be described using the tight-binding approximation.

$$H = -t \left(\sum_{\langle i,j \rangle} |i\rangle \langle j| + H.C \right) + \sum_i V_i |i\rangle \langle i| \quad (3.1)$$

where i and j are the nearest neighbor sited in honeycomb lattice, t_{ij} is the hopping parameter and V_i is the onsite potential. This serves as the model hamiltonian which is used to calculate the connection matrix elements (hopping terms) and isolated slice energies (for the conductor) in the RGF scheme.

3.1.2 Slicing schemes for graphene

The RGF method enables computation of $G_{i,j}$ for a large variety of settings. For instance, the graphene sheet can have an arbitrary number of layers and different edge orientations: zigzag, armchair, and chiral. Also, the tight-binding model can include next-nearest neighbor hopping terms in addition to the nearest neighbor ones. These elements have to be taken into account when choosing the slice unit cell employed by the recursive method.

We limit our discussion to a single layer of graphene with only nearest neighbour in the model hamiltonian. Some of the slicing schemes are as follows.

Armchair lattice

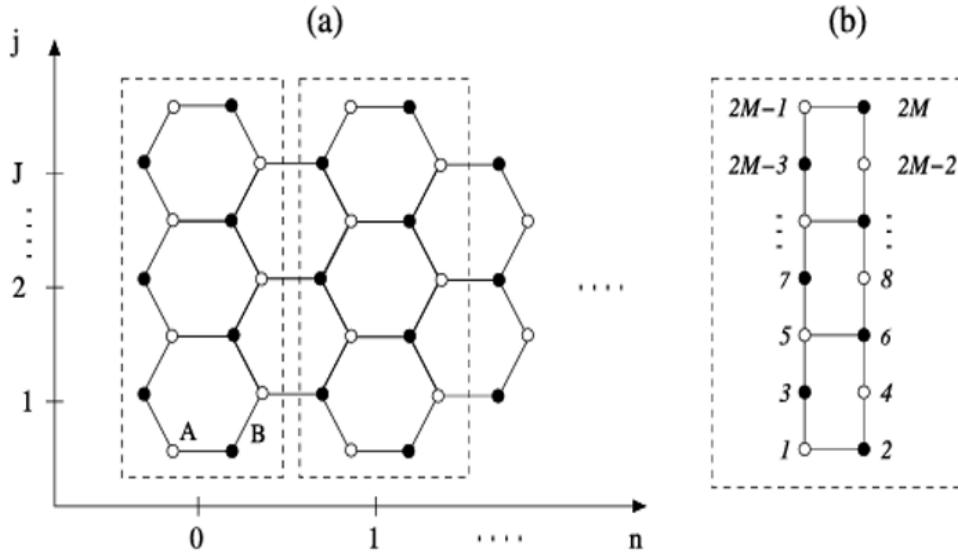
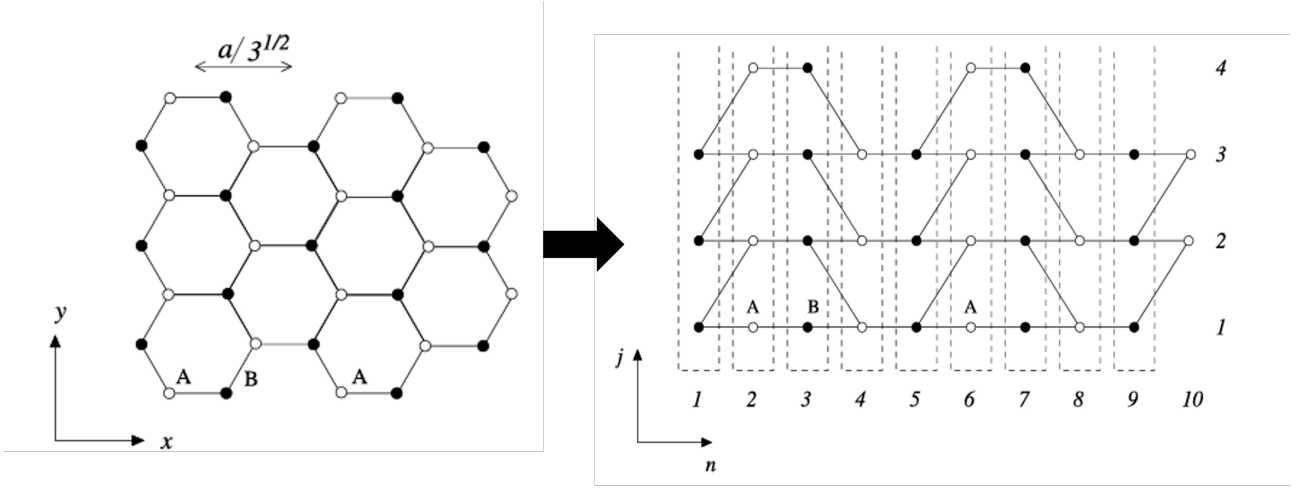


Fig. 3.2: (a) Structure of the honeycomb armchair lead. Note that the number of interconnect channels runs from 1 to J , while the number of vertical coordinate points in the slice is $M = 2J + 1$. The slices for decimation are defined by the boxed regions. (b) Internal index structure of the elementary slice

Fig 3.2 shows a direct slicing scheme for honeycomb lattice of graphene with armchair edge orientation, with the individual slice. Here, (and for the following slicing schematics) atoms from different sublattices are represented by empty(B) and full(A) circles. This scheme involves a large number of sites in each slice along with hopping terms within the slice, and thus is computationally intensive. Consider another scheme for the same lattice configuration as shown in Fig 3.3.



Efficient way to slice an armchair graphene ribbon ("pine tree" configuration)

Fig. 3.3

This scheme involves bending the original lattice configuration such that efficient slicing is obtained while retaining the structural features of the original configuration. It is much better in terms of sites per slice and only involves two inter-slice hopping terms per site in the slice (no next-to-nearest neighbor hopping), hence it is much more efficient. The relation between the lattice size and the graphene sheet dimensions are:

$$\frac{L}{a_0} = \frac{\sqrt{3}}{2} \left(\frac{N}{2} - 1 \right) + \frac{\sqrt{3}}{6} \quad (3.2)$$

and

$$\frac{W}{a_0} = M - 1 \quad (3.3)$$

Zigzag edges

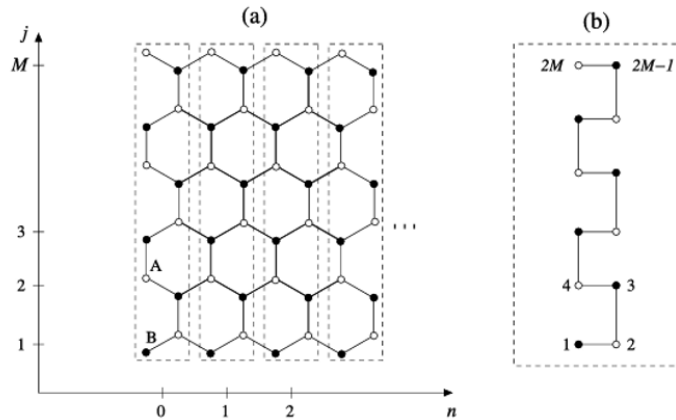


Fig. 3.4: (a) Structure of the honeycomb zigzag lead. Note that the number of interconnect channels runs from 1 to $J = M$, where M is the number of vertical coordinate points in the graphene. The slices for decimation are defined by the boxed regions. (b) Internal index structure of the elementary slice

Fig 3.4 shows a direct slicing scheme for honeycomb lattice of graphene with zigzag edge orientation, with the individual slice. This scheme involves a large number of sites in each slice along with hopping terms within the slice, and thus is computationally intensive. Consider another scheme for the same lattice configuration as shown in Fig 3.5.

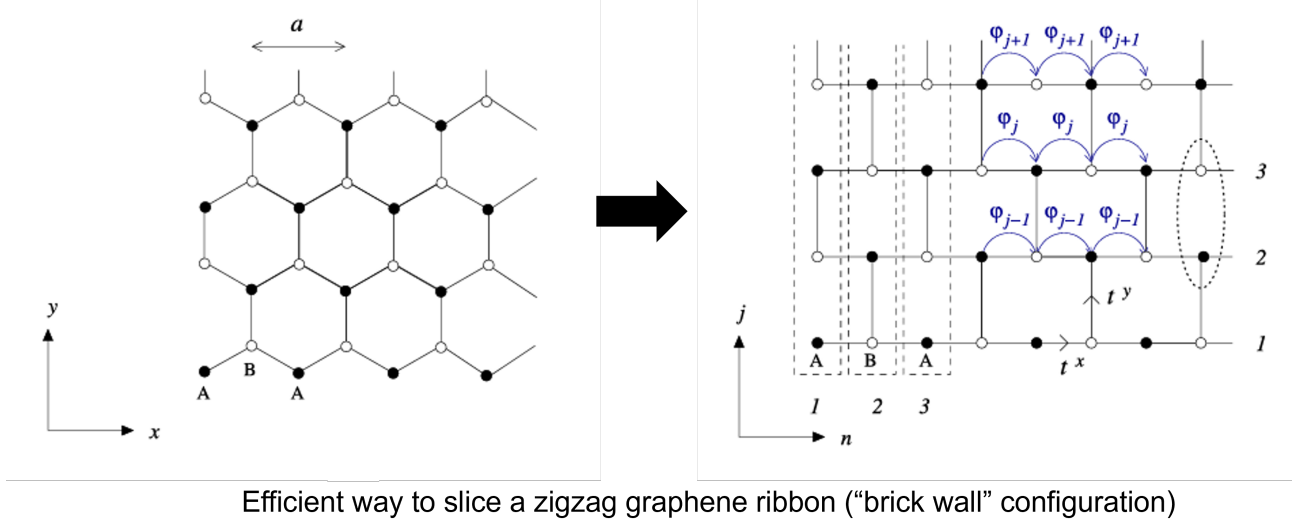


Fig. 3.5

Again, the original lattice configuration is bent such that more efficient slicing is obtained while retaining the structural features of the original configuration. For odd and even number of sites per slice we have isolated sites (at the bottom and top of the slice respectively) and dimers. It is still better in terms of sites per slice and improves upon relevant hopping terms per site. The relation between the lattice size and the graphene sheet dimensions are:

$$\frac{W}{a_0} = \frac{\sqrt{3}}{2} (M - 1) + \frac{\sqrt{3}}{6} \quad (3.4)$$

and

$$\frac{L}{a_0} = \frac{N - 1}{2} \quad (3.5)$$

3.2 Results

Use of efficient slicing scheme enables easier calculation of the relevant Green's functions using the RGF method, which are then used to calculate relevant physical quantities as follows.

3.2.1 Calculating observables

We have,

$$H = -t \left(\sum_{\langle i,j \rangle} |i\rangle \langle j| + H.C \right) + \sum_i V_i |i\rangle \langle i| \quad (3.6)$$

The discretized Green's function matrix is given by,

$$\begin{bmatrix} G_p & G_{pC} \\ G_{Cp} & G_C \end{bmatrix} = \begin{bmatrix} (E + i\eta)I - H_p & u \\ u^\dagger & EI - H_C \end{bmatrix}^{-1} \quad (3.7)$$

Here, G_C is the Green's function matrix of the graphene sample, G_P is the Green's function of the lead slice in contact with the sample, and u is the connection matrix between the two components.

The corresponding surface self-energies of the leads are defined as,

$$\begin{aligned}\Sigma_L^r(E) &= u_L g_L^r(E) u_L^\dagger \\ \Sigma_R^r(E) &= u_L^\dagger g_R^r(E) u_L\end{aligned}\tag{3.8}$$

Here, g_L^r and g_R^r are the Green's function at the surface of the leads when they are decoupled from the system.

$$g_{L,R}^r(E) = (E + i0^+ - h_{L,R})^{-1}\tag{3.9}$$

Using 1.61, we write,

$$T = \text{Tr}_s[\Gamma_L G_{LR}^r \Gamma_R G_{RL}^a]\tag{3.10}$$

Here, T is the transmission probability of the electron going from left lead to right and trace is carried over all sites. Γ here are called level widths, and are defined as (using 1.62),

$$\Gamma_p(i, i') = \sum_a \chi_a(i) \frac{\hbar v_a}{a_0} \chi_a(i')\tag{3.11}$$

Alternatively (using 1.63),

$$\Gamma_{L,R} = i[\Sigma_{L,R}^r(E) - \Sigma_{L,R}^{r\dagger}(E)]\tag{3.12}$$

Using RGF we write,

$$\mathbf{T} = \text{Tr}_s[\mathbf{\Gamma}_L \mathbf{G}_{0,N+1}^r(\mathbf{E}) \mathbf{\Gamma}_R \mathbf{G}_{N+1,0}^a(\mathbf{E})]\tag{3.13}$$

Spectral function: We can define the discretized spectral function as (using 1.65),

$$A(n, n'; E) \equiv i[G_{n,n'}^R - G_{n,n'}^A] = G_{n,n'}^R \Gamma G_{n,n'}^A = G_{n,n'}^A \Gamma G_{n,n'}^R\tag{3.14}$$

Thus, the local density of states is defined by (using 1.69),

$$\rho(n, n'; E) = \frac{1}{2\pi} A(n, n'; E) = -\frac{1}{\pi} \text{Im}[G(n, n'; E)]\tag{3.15}$$

3.2.2 Numerical results

In this section, some numerical results for ballistic transport in disorder-free samples are presented.

Fig. 3.6 shows the conductance as a function of the fermi energy in the contacts. The results are for a graphen sample (short single ribbon) with armchair edges, $M = 360$ and $N = 70$, in the clean limit (no bulk or edge disorder). We note excellent agreement for large systems, and worse for smaller systems. A strong asymmetry and a lack of well-defined oscillations occurs if the system is not large enough (not shown).

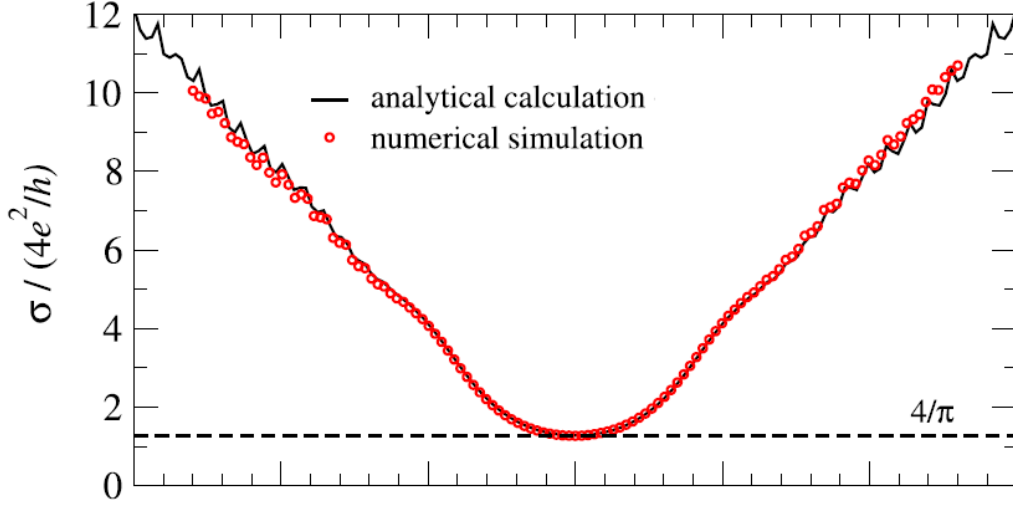


Fig. 3.6: Conductance vs Fermi energy in contact. Solid line: analytic result and red data points: numerical results. ($M = 360, N = 70$)

Fig. 3.7 shows the plot of conductance for different aspect ratios for the same graphen sample (in clean limit, with different lead configurations). For all the curves, $M = 120$.

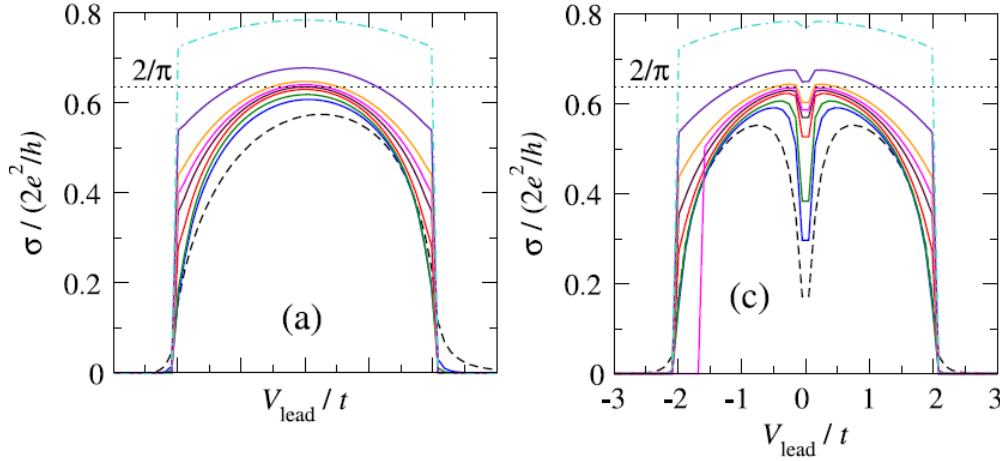


Fig. 3.7: Conductance vs Fermi energy in contact for different aspect ratios $M = 360, N = 12$ (dashed line), 24, 36, 72, 96, 108, 120, 148, and 200. Plot (a) is for square lattice contact. Plot (b) is for honeycomb contact.

There is clear increase (decrease) in conductance (resistance) observed with increase in N (and the number of modes). This is in agreement with our analysis of contact resistance in ballistic conductors.

Chapter 4

Comments and Conclusion

The numerical results suggest excellent usability of the RGF method for computation of Green's function and associated transport quantities (in a two-probe setup) for large ballistic conductor systems (only one graphene ribbon considered). The accuracy degrades for smaller systems (which include significant contributions from quantum interference effects and evanescent modes).

However, the method is robust and allows consideration of an arbitrary number of layers and different edge orientations. For multiple layers, there may be more efficient slicing schemes than those described in our work. With increase in the number of effective sites, the recursive calculation of Green's functions become more difficult and computationally intensive.

Our transport analysis only considered non-interacting electrons (no electron-electron, or electron-phonon interactions) along with no disorder (bulk or edge) in the system. These interactions can cause phase destruction of the electronic paths, this process is called *dephasing*. Such systems can also be studied using this method (by a suitable choice of hopping parameter and local potential).

Lastly, this method can be extended to incorporate non-equilibrium Green's functions and systems with finite temperature.

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